

EROLLA RISHVIN REDDY

Cybersecurity Engineer • Digital Forensics • IoT Security

Network Security • Digital Forensics • Python • Linux • OWASP • PCAP Analysis • MQTT • Security Automation

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Professional Summary

Computer Science Engineering student specializing in **Blockchain, IoT & Cybersecurity (BIC)** with a focused interest in **Cybersecurity, Security Engineering, Digital Forensics, Network Security, IoT Security, and Security Operations**. Experienced in developing security-focused systems for **network traffic analysis, PCAP investigation, digital evidence preservation, vulnerability assessment, security automation, biometric authentication, and secure IoT architectures** using **Python, TypeScript, Linux, MQTT, Docker, PostgreSQL, OWASP principles, cryptographic hashing, and REST APIs**. Founder of **Rishvin Labs**, developing practical security tools and platforms including network forensics workbenches, IoT forensic evidence systems, browser-native email forensics, and AI-assisted security operations. Hands-on experience with **threat modeling, security monitoring, vulnerability analysis, authentication, evidence integrity, incident investigation, and secure software development**, supported by a strong foundation in **computer networks, operating systems, data structures, database systems, and cybersecurity**.

Education

Woxsen University

B.Tech in Computer Science and Engineering

Specialization: Blockchain, IoT & Cybersecurity (BIC)

Hyderabad, India

2024 – 2028

CGPA: **9.01/10**

Cybersecurity & Security Engineering Skills

Cybersecurity: Cybersecurity, Information Security, Security Engineering, Application Security, Network Security, IoT Security, Secure Software Development, Security Architecture, Threat Modeling

Digital Forensics: Digital Forensics, Network Forensics, Email Forensics, PCAP Analysis, Packet Analysis, Evidence Preservation, Chain of Custody, Timeline Reconstruction, Evidence Integrity, Cryptographic Hashing, SHA-256

Network Security: TCP/IP, HTTP/HTTPS, DNS, MQTT, Network Traffic Analysis, Protocol Analysis, Packet Inspection, Network Monitoring, Traffic Filtering, Wireless Security

Web Application Security: OWASP, Web Security Testing, Vulnerability Assessment, Cross-Site Scripting (XSS), Authentication Security, Authorization, Input Validation, Session Security, Burp Suite, PortSwigger

Security Operations: Security Operations Center (SOC), Security Monitoring, Threat Intelligence, Vulnerability Management, Risk Assessment, Risk Scoring, Incident Analysis, Security Automation, Automated Security Reporting

Identity & Access Security: Biometric Authentication, Identity Verification, Access Control, Authentication Workflows, Secure Credential Handling, Duplicate Identity Detection, Auditability

Cryptography & Security Protocols: SHA-256, AES-256-GCM, Cryptographic Hashing, Data Integrity, Secure Communication, Encryption, Integrity Verification, Blockchain-Based Evidence Integrity

Security Development: Python, TypeScript, JavaScript, Node.js, FastAPI, PostgreSQL, REST APIs, Docker, Linux, Git, GitHub

Security Tools & Platforms: Burp Suite, PortSwigger Web Security Academy, Wireshark, OWASP, Postman, Docker, Linux, GitHub, n8n

Security Engineering Fundamentals: Secure System Design, Incident Response Concepts, Vulnerability Analysis, Risk Management, Computer Networks, Operating Systems, Database Security, Software Engineering, Problem Solving

Technical Experience

Founder & Security Engineer – Rishvin Labs

2025 – Present

- Founded and lead **Rishvin Labs**, developing security-focused software across **cybersecurity, digital forensics, network security, IoT security, security automation, secure authentication, and blockchain-based evidence integrity**.
- Architected and developed security platforms for **network traffic analysis, PCAP investigation, email forensics, IoT security monitoring, vulnerability assessment, biometric authentication, and forensic evidence preservation**.
- Implemented security-oriented systems using **Python, TypeScript, Node.js, PostgreSQL, Docker, MQTT, REST APIs, cryptographic hashing, and blockchain technologies**.
- Applied **threat modeling, secure system design, authentication, authorization, data integrity, vulnerability analysis, security monitoring, and forensic investigation** across independent engineering projects.

- Built security automation workflows integrating **n8n**, **vulnerability analysis**, **threat intelligence**, **AI-assisted security analysis**, **risk scoring**, **automated reporting**, and **event-driven security operations**.

Pega Platform Intern – Pegasystems

Aug 2026 – Present

- Selected for the **National Internship Program 2026 with SmartBridge**, gaining hands-on exposure to enterprise application development and workflow automation using the **Pega Platform**.
- Developing practical knowledge of **Business Process Management (BPM)**, **Case Management**, **Digital Process Automation (DPA)**, **enterprise workflows**, **application configuration**, and **data integration**.
- Applying structured enterprise software practices involving **application lifecycle management**, **business process modeling**, **case lifecycle design**, **workflow orchestration**, and **secure enterprise application development**.

Cybersecurity & Security Engineering – Academic & Independent Projects

2024 – Present

- Designed and implemented security systems spanning **network forensics**, **digital evidence analysis**, **IoT security**, **web application security**, **biometric authentication**, **vulnerability assessment**, and **security operations automation**.
- Performed practical security analysis involving **network traffic inspection**, **PCAP analysis**, **HTTP/HTTPS traffic**, **vulnerability testing**, **XSS labs**, **authentication workflows**, **cryptographic verification**, and **forensic investigation**.
- Applied **secure software development**, **threat modeling**, **cryptographic hashing**, **access control**, **evidence integrity**, **incident analysis**, and **security monitoring** to real-world-inspired engineering problems.
- Integrated cybersecurity with **IoT**, **blockchain**, **AI**, **digital forensics**, and **full-stack engineering** to develop multidisciplinary security solutions and investigation platforms.

Featured Security Projects

NetInspect – Network Traffic Analysis & Digital Forensics Workbench

Python, JavaScript, PCAP, Browser APIs, Network Security

- Built a **browser-based network forensics workbench** for local PCAP inspection, packet analysis, protocol investigation, and security-focused network traffic analysis.
- Implemented **packet inspection**, **protocol analysis**, **traffic filtering**, **search**, **HTTP/DNS analysis**, and **investigation workflows** for analyzing network communications and suspicious traffic.
- Designed a **privacy-first local processing architecture** that enables forensic analysis without requiring sensitive packet captures to be uploaded to a remote server.

ChainForensics – IoT Security & Digital Forensic Evidence Platform

TypeScript, Node.js, MQTT, Ethereum, Solidity, IPFS, SHA-256

- Engineered an **IoT security and digital forensics platform** for collecting, verifying, preserving, and investigating security events generated by connected devices.
- Implemented an evidence pipeline using **MQTT telemetry**, **deterministic JSON serialization**, **SHA-256 hashing**, **Ethereum smart contracts**, and **IPFS** to establish verifiable evidence integrity.
- Designed mechanisms for **tamper detection**, **immutable evidence registration**, **chain-of-custody verification**, and **forensic timeline reconstruction** across distributed IoT environments.

AI Security Guardian – AI-Powered Security Operations Platform

Python, n8n, PostgreSQL, Docker, OWASP, LLM APIs

- Architected an **AI-assisted Security Operations Center (SOC)** platform for automated attack surface management, vulnerability assessment, threat intelligence, security analysis, and reporting.
- Integrated **vulnerability scanning**, **deterministic risk scoring**, **AI-assisted security analysis**, **threat intelligence enrichment**, **monitoring**, and **automated security workflows** using n8n.
- Designed a containerized security architecture using **Python**, **PostgreSQL**, **Docker**, and **event-driven automation** for repeatable vulnerability analysis and security operations.

MBOX Viewer – Browser-Native Email Forensics Tool

TypeScript, JavaScript, HTML5, CSS3, Browser APIs, Digital Forensics

- Developed a **zero-backend email forensics application** for locally parsing, searching, filtering, and investigating large MBOX email archives.
- Implemented **email metadata extraction**, **attachment analysis**, **advanced search**, **filtering**, and **forensic investigation workflows** for structured examination of email evidence.
- Designed a privacy-preserving architecture that keeps sensitive email archives **client-side**, eliminating the need for server-side forensic processing.

Biometric Voting System – Secure Identity & Authentication Platform

TypeScript, React, Node.js, ESP32, Fingerprint Biometrics

- Developing a **secure biometric authentication platform** using fingerprint-based identity verification for controlled and auditable voter authentication.

- Upgraded the prototype from an **R307 fingerprint sensor to a Mantra fingerprint device**, strengthening hardware integration and real-world biometric verification workflows.
- Designed security controls for **identity verification, duplicate-vote prevention, access control, auditability, and secure hardware-software integration**.

Web Application Security Testing – PortSwigger Web Security Academy

Burp Suite, HTTP, Web Security, OWASP, XSS

- Performed hands-on **web application security testing** through PortSwigger Web Security Academy labs, analyzing HTTP requests, responses, application behavior, and security vulnerabilities.
- Practiced identifying and exploiting controlled **Cross-Site Scripting (XSS)** vulnerabilities using **Burp Suite** and structured web-security testing methodologies.
- Applied foundational **OWASP web security principles, request manipulation, input analysis, and vulnerability assessment** in controlled security-training environments.

Certifications

Introduction to Cyber Attacks – New York University (Coursera)

Career Essentials in GitHub Professional Certificate – GitHub & LinkedIn Learning

Practical GitHub Actions – LinkedIn Learning

Practical GitHub Code Search – LinkedIn Learning

Practical GitHub Project Management and Collaboration – LinkedIn Learning

Programming for Everybody (Getting Started with Python) – University of Michigan (Coursera)

Data Structures Using Python: An Introduction – Coursera

Object-Oriented Design – University of Alberta (Coursera)

Java Programming – Coursera

Relevant Coursework

Cybersecurity & Networking: Cybersecurity, Computer Networks, Network Security, Internet Technologies, IoT Security, Secure Communication, Digital Forensics

Computer Science: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Software Engineering

Security Engineering: Information Security, Secure Software Development, Authentication, System Design, Database Security, Threat Modeling

Supporting Foundations: Probability & Statistics, Blockchain Technology, Internet of Things, Technical Writing, Problem Solving

Patent & Innovation

Government of India Registered Design Patent – IoT Connectivity Device *Co-Inventor — Design No: 470097-001*

- Co-inventor of a **Government of India registered design patent** for an IoT connectivity device, demonstrating applied innovation across **IoT security, connected-device engineering, secure communication, and smart infrastructure**.
- Contributed to an engineering solution involving **device connectivity, communication workflows, and connected-system integration**, translating an engineering concept into a formally registered intellectual property asset.